A2 · Unit 4 · WCH14 Rates, Equilibria and Further Organic Chemistry

Acid-base equilibria

Acids and bases are all about protons moving from one particle to another. This topic turns that idea into numbers: the pH scale, the ionic product of water, the acid dissociation constant K_a, the shapes of titration curves, and the buffers that keep the pH of your blood almost perfectly steady.

Brønsted–Lowry acids and bases

- **A Brønsted–Lowry acid** is a proton (H^+) donor.
- **A Brønsted–Lowry base** is a proton acceptor. It needs a lone pair of electrons to bond to the proton.

An acid can only give away a proton if something accepts it, so acid–base reactions always involve two pairs. When an acid loses its proton, what is left can accept a proton back: it is the acid's **conjugate base**. Likewise a base that gains a proton becomes its **conjugate acid**. A conjugate pair differs by exactly one H^+.

Reaction	Acid 1	Base 2	Acid 2	Base 1
$HCl + H_2O \rightarrow H_3O^+ + Cl^-$	HCl	H_2O	H_3O^+	Cl^-
$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$	H_2O	NH_3	NH_4^+	OH^-
$HNO_3 + HNO_2 \rightleftharpoons NO_3^- + H_2NO_2^+$	HNO_3	HNO_2	$H_2NO_2^+$	NO_3^-

Water acts as a base towards HCl but as an acid towards NH_3: a substance that can do both is called **amphoteric**. In the last example nitrous acid is forced to act as a base, because nitric acid is the stronger acid; when two acids are mixed, the one with the larger K_a donates the proton.

pH and strong acids

$$\mathrm{pH} = -\log_{10}[\mathrm{H}^+] \qquad [\mathrm{H}^+] = 10^{-\mathrm{pH}}$$

The pH scale is logarithmic so that hydrogen-ion concentrations ranging over many powers of ten fit onto a scale of roughly 0 to 14. Each fall of one pH unit means ten times more H^+. Give pH values to two decimal places.

A **strong acid** is fully dissociated in water, so for a monoprotic strong acid such as HCl or HNO_3, $[H^+]$ equals the concentration of the acid.

pH of a strong acid, and back again

1. 0.0250 mol dm^{-3} nitric acid: [H$^+$] = 0.0250 mol dm^{-3}, so pH = −log 0.0250 = **1.60**.

2. A hydrochloric acid solution has pH 2.45: [H$^+$] = $10^{-2.45}$ = 3.55 × 10^{-3} mol dm^{-3}, and that is also the concentration of the acid.

The ionic product of water, K_w

Water itself ionises very slightly: $H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$. So little water ionises that $[H_2O]$ is effectively constant, and it is folded into the constant to give the **ionic product of water**:

$$K_w = [\mathrm{H^+}][\mathrm{OH^-}] = 1.00 \times 10^{-14}\ \mathrm{mol^2\,dm^{-6}}\ \text{at 298 K} \qquad \mathrm{p}K_w = -\log K_w = 14.00$$

K_w applies to every aqueous solution, so if you know [OH$^-$] you can find [H$^+$], and the other way round.

Neutral water is not always pH 7

A solution is **neutral** when [H$^+$] = [OH$^-$]. At 298 K that gives [H$^+$] = $\sqrt{(1.00 \times 10^{-14})}$ = 1.00 × 10^{-7} mol dm^{-3} and pH 7.00. The ionisation of water is **endothermic**, so warming it increases K_w. At 40 °C, K_w = 2.92 × 10^{-14} mol^2 dm^{-6}, so [H$^+$] = $\sqrt{(2.92 \times 10^{-14})}$ = 1.71 × 10^{-7} mol dm^{-3} and pH = 6.77. That water is still neutral, because [H$^+$] and [OH$^-$] are still equal; neutral simply has a lower pH at higher temperatures.

pH of strong bases

pH of a strong base

1. Strong bases such as NaOH are fully dissociated, so 0.0500 mol dm^{-3} NaOH has [OH$^-$] = 0.0500 mol dm^{-3}.

2. Use K_w: $[\mathrm{H^+}] = \dfrac{K_w}{[\mathrm{OH^-}]} = \dfrac{1.00 \times 10^{-14}}{0.0500} = 2.00 \times 10^{-13}\ \mathrm{mol\,dm^{-3}}$.

3. pH = −log(2.00 × 10^{-13}) = **12.70**.

Watch for bases that release two hydroxide ions: 0.0200 mol dm^{-3} $Ba(OH)_2$ gives [OH$^-$] = 0.0400 mol dm^{-3} and pH 12.60.

Weak acids and K_a

A **weak acid** is only slightly dissociated: the solution is an equilibrium mixture that is mostly undissociated HA. Its strength is measured by the **acid dissociation constant**, K_a, which is simply K_c for the dissociation:

$$\mathrm{HA(aq)} \rightleftharpoons \mathrm{H^+(aq)} + \mathrm{A^-(aq)} \qquad K_a = \frac{[\mathrm{H^+}][\mathrm{A^-}]}{[\mathrm{HA}]} \qquad \mathrm{p}K_a = -\log K_a$$

- The units of K_a are mol dm^{-3}.
- The **larger K_a** (the smaller pK_a), the further the equilibrium lies to the right and the stronger the acid. Ethanoic acid has $K_a = 1.7 \times 10^{-5}$ mol dm^{-3}, pK_a = 4.77.
- **Strong and weak are not the same as concentrated and dilute.** Strength is about how much the acid dissociates; concentration is about how much acid is in each dm^3.

Calculating the pH of a weak acid

Two approximations make the calculation simple:

1. [H$^+$] = [A$^-$], because each HA that dissociates gives one of each. (This ignores the tiny amount of H$^+$ from the water.)
2. [HA] at equilibrium ≈ the concentration of acid you started with, because so little of it dissociates.

Then $K_a = \dfrac{[\mathrm{H^+}]^2}{[\mathrm{HA}]_{\text{initial}}}$, so $[\mathrm{H^+}] = \sqrt{K_a \times [\mathrm{HA}]_{\text{initial}}}$. The second approximation breaks down for acids that are fairly strong (large K_a) or very dilute, because then a significant fraction of the acid dissociates.

pH of 0.0500 mol dm^{−3} ethanoic acid

1. $[\mathrm{H^+}]^2 = K_a \times [\mathrm{HA}] = 1.7 \times 10^{-5} \times 0.0500 = 8.5 \times 10^{-7}$
2. $[\mathrm{H^+}] = \sqrt{8.5 \times 10^{-7}} = 9.22 \times 10^{-4}\ \mathrm{mol\,dm^{-3}}$
3. pH = −log(9.22 × 10^{-4}) = **3.04**.

A strong acid of the same concentration would have pH 1.30. Only about 2% of the ethanoic acid molecules are dissociated.

K_a from a measured pH

1. 0.200 mol dm^{-3} propanoic acid has pH 2.79. Find K_a and pK_a.

2. $[H^+] = 10^{-2.79} = 1.62 \times 10^{-3}$ mol dm^{-3} = $[A^-]$.

3. $K_a = \dfrac{(1.62 \times 10^{-3})^2}{0.200} = 1.32 \times 10^{-5}\ \text{mol dm}^{-3}$, so $pK_a = 4.88$.

Diluting acids

Diluting a strong acid ten times divides $[H^+]$ by ten, so the pH rises by exactly 1. Diluting a weak acid ten times raises its pH by only about 0.5: 0.0500 mol dm^{-3} ethanoic acid has pH 3.04, but 0.00500 mol dm^{-3} has pH 3.54. Adding water shifts the dissociation equilibrium to the right, so a larger fraction of the acid dissociates and partly makes up the lost H^+. Mathematically, $[H^+]$ is proportional to the square root of the concentration.

KEY IDEA

Core practical: finding K_a of a weak acid. Calibrate a pH meter with buffers of known pH. Titrate 25.0 cm^3 of the weak acid with sodium hydroxide to find the volume needed for neutralisation. Then add exactly half that volume of alkali to a fresh 25.0 cm^3 of acid and measure the pH. At half-neutralisation $[HA] = [A^-]$, so pH = pK_a (explained below).

Titration curves

A titration curve plots pH against the volume of one solution added to the other. To record one, put a known volume of acid in a beaker with a calibrated pH meter, add the alkali from a burette a little at a time, stirring and recording the pH after each addition, with smaller additions near the end point. The **equivalence point** is where the acid and base have reacted exactly in the ratio of the equation; on the graph it is the middle of the steep, nearly vertical part.

Titration (base added to 0.1 mol dm^{-3} acid)	Starting pH	Steep section	pH at equivalence
strong acid + strong base (HCl + NaOH)	about 1	about 3 to 11	7
weak acid + strong base (CH_3COOH + NaOH)	about 3	about 7 to 11	above 7 (about 8.7)
strong acid + weak base (HCl + NH_3)	about 1	about 3 to 7	below 7 (about 5.3)
weak acid + weak base (CH_3COOH + NH_3)	about 3	no steep section	about 7, but hard to pin down

The pH at equivalence is not always 7 because the salt formed can itself react with water. Sodium ethanoate contains ethanoate ions, the conjugate base of a weak acid, which take protons from water and leave OH^-: the solution is alkaline. Ammonium chloride contains NH_4^+, a weak acid, so its solution is acidic.

Half-neutralisation: pH = pK_a

On the curve for a weak acid with a strong base, the pH rises quickly at first and then flattens into a **buffer region**. Halfway to the equivalence point, exactly half of the acid has been turned into its salt, so [HA] = [A^-]. These cancel in the K_a expression, leaving $K_a = [H^+]$, so **pH = pK_a**. This is the basis of the core practical above.

Choosing an indicator

An indicator is itself a weak acid, HIn, whose undissociated form and conjugate base have different colours: HIn ⇌ $H^+ + In^-$. Adding acid pushes the equilibrium left, giving the colour of HIn; adding alkali removes H^+ and pulls it right, giving the colour of In^-. The **end point**, where the colour changes, is when [HIn] = [In^-]; it happens over a range of about 2 pH units centred on the indicator's pK_{in}.

A suitable indicator changes colour completely within the **steep part** of the titration curve, so that one drop of titrant takes it all the way through its colour change at the equivalence point.

Indicator	Colour in acid → alkali	pH range	Use for
methyl orange	red → yellow (orange at the end point)	about 3.1 to 4.4	strong acid with strong or weak base
phenolphthalein	colourless → pink	about 8.3 to 10.0	strong or weak acid with strong base

For a weak acid with a weak base there is no steep section, so no indicator gives a sharp end point; use a pH meter instead.

Diprotic acids such as ethanedioic acid, $H_2C_2O_4$, lose their two protons one at a time, so their curve with sodium hydroxide has two steep sections and two equivalence points. In the same way, adding acid to sodium carbonate gives two steps: $CO_3^{2-} + H^+ \rightarrow HCO_3^-$, then $HCO_3^- + H^+ \rightarrow H_2O + CO_2$.

Buffer solutions

A **buffer** is a solution whose pH changes very little when small amounts of acid or alkali are added, or when it is diluted.

- **Acidic buffer**: a weak acid and a salt of that acid, for example ethanoic acid and sodium ethanoate. It holds a pH below 7.
- **Basic buffer**: a weak base and a salt of that base, for example ammonia and ammonium chloride. It holds a pH above 7.

How an acidic buffer works

The buffer contains large reservoirs of both the weak acid, CH_3COOH, and its conjugate base, CH_3COO^-, which comes almost entirely from the salt: $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$.

- **Add a little acid**: the added H^+ combines with the large supply of ethanoate ions, $CH_3COO^- + H^+ \rightarrow CH_3COOH$, so the equilibrium shifts left and almost all the added H^+ is removed.
- **Add a little alkali**: the OH^- reacts with H^+ to make water, and the equilibrium shifts right as more ethanoic acid dissociates to replace it. The OH^- is also removed directly: $CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$.
- Either way the ratio $[CH_3COOH] / [CH_3COO^-]$ changes only slightly, because both reservoirs are large, so $[H^+]$ and the pH barely change.

Calculating the pH of a buffer

Rearranging K_a: $[\mathrm{H^+}] = K_a \times \dfrac{[\mathrm{HA}]}{[\mathrm{A^-}]}$. Take [HA] as the acid added and $[A^-]$ as the salt added; the tiny amounts that dissociate or react with water make no noticeable difference. Because acid and salt share the same final volume, you can use their **moles** directly in the ratio.

Buffer made by mixing acid and salt

1. 50.0 cm^3 of 0.200 mol dm^{-3} ethanoic acid is mixed with 50.0 cm^3 of 0.100 mol dm^{-3} sodium ethanoate. $K_a = 1.7 \times 10^{-5}$ mol dm^{-3}.
2. Moles of acid = 0.0500 × 0.200 = 0.0100 mol. Moles of ethanoate = 0.0500 × 0.100 = 0.00500 mol.
3. $[\mathrm{H^+}] = 1.7 \times 10^{-5} \times \dfrac{0.0100}{0.00500} = 3.4 \times 10^{-5}\ \mathrm{mol\,dm^{-3}}$, so pH = **4.47**.

Buffer made by partly neutralising a weak acid

1. 50.0 cm^3 of 0.400 mol dm^{-3} ethanoic acid is mixed with 30.0 cm^3 of 0.200 mol dm^{-3} NaOH. Find the pH.
2. Moles of acid = 0.0200 mol; moles of NaOH = 0.00600 mol. The NaOH reacts completely: $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$.
3. Left over: 0.0200 − 0.00600 = 0.0140 mol of acid; formed: 0.00600 mol of ethanoate.

4. $[H^+] = 1.7 \times 10^{-5} \times \frac{0.0140}{0.00600} = 3.97 \times 10^{-5}\ \text{mol dm}^{-3}$, so pH = **4.40**.

Adding acid to a buffer

1. 1.00 dm^3 of buffer contains 0.100 mol of ethanoic acid and 0.100 mol of sodium ethanoate, so its pH = pK_a = 4.77. Now 0.0100 mol of HCl is added.
2. The H^+ turns ethanoate into ethanoic acid: acid becomes 0.110 mol, ethanoate becomes 0.0900 mol.
3. $[H^+] = 1.7 \times 10^{-5} \times \frac{0.110}{0.0900} = 2.08 \times 10^{-5}\ \text{mol dm}^{-3}$, so pH = **4.68**, a fall of only 0.09.

The same acid added to 1.00 dm^3 of pure water would take the pH from 7.00 to 2.00.

Buffers in blood

Blood plasma must stay between pH 7.35 and 7.45. The main buffer is carbonic acid and hydrogencarbonate ions: $H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$. Extra acid from metabolism is taken up by HCO_3^-, shifting the equilibrium left; extra alkali removes H^+, and more H_2CO_3 dissociates to replace it. The carbonic acid level is itself controlled by breathing out carbon dioxide ($H_2CO_3 \rightleftharpoons H_2O + CO_2$). Buffers are also used in foods, shampoos and medicines to keep the pH where the product works best.

Enthalpy change of neutralisation

The **standard enthalpy change of neutralisation** is the enthalpy change when an acid and an alkali react under standard conditions to form one mole of water. For any strong acid with any strong alkali it is almost the same, about −57 kJ mol^{-1}, because both are fully ionised and the only reaction is $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$. With a **weak** acid (or weak base) it is slightly less exothermic, for example about −56 kJ mol^{-1} for ethanoic acid, because some energy is used up dissociating the acid as the reaction proceeds.

Quick checks

QUICK CHECK

In $HSO_4^- + H_2O \rightleftharpoons SO_4^{2-} + H_3O^+$, which is the conjugate base of HSO_4^-?

- H_2O
- H_3O^+
- SO_4^{2-}
- H_2SO_4

HSO_4^- donates a proton and becomes SO_4^{2-}; the pair differs by one H^+. H_3O^+ is the conjugate acid of water.

QUICK CHECK

What is the pH of 0.0150 mol dm^{-3} potassium hydroxide at 298 K?

- 1.82
- 12.18
- 11.82
- 13.18

$[H^+] = 1.00 \times 10^{-14} \div 0.0150 = 6.67 \times 10^{-13}$ mol dm^{-3}, so pH = 12.18. 1.82 is the pOH, which is what you get if you take $-\log[OH^-]$ and stop.

QUICK CHECK

Pure water at 60 °C has pH 6.51. Which statement is correct?

- The water is slightly acidic
- K_w is smaller than at 25 °C
- $[H^+]$ is greater than $[OH^-]$
- The water is neutral and K_w is larger than at 25 °C

In pure water $[H^+]$ always equals $[OH^-]$, so it is neutral. Ionisation of water is endothermic, so K_w rises with temperature and the neutral pH falls below 7.

QUICK CHECK

What is the pH of 0.100 mol dm^{-3} of a weak acid with $K_a = 1.0 \times 10^{-5}$ mol dm^{-3}?

- 3.00
- 1.00
- 6.00
- 5.00

$[H^+] = \sqrt{(1.0 \times 10^{-5} \times 0.100)} = 1.0 \times 10^{-3}$ mol dm^{-3}, so pH = 3.00. 1.00 would be right only for a strong acid. 6.00 comes from forgetting the square root.

QUICK CHECK

Which indicator is suitable for titrating ammonia solution with hydrochloric acid?

- phenolphthalein, because the equivalence point is above 7
- methyl orange, because the steep section is roughly pH 3 to 7
- either, because both change colour near pH 7
- neither, because there is no steep section

A strong acid with a weak base has its steep section in the acidic region and an equivalence point near pH 5, which suits methyl orange. Phenolphthalein would change colour long before the equivalence point.

QUICK CHECK

25.0 cm^3 of a weak acid needs 24.0 cm^3 of NaOH for neutralisation. After 12.0 cm^3 of NaOH has been added the pH is 4.20. What is K_a?

- 4.20 mol dm^{-3}
- 1.6×10^{-2} mol dm^{-3}
- 6.3×10^{-5} mol dm^{-3}
- 4.0×10^{-9} mol dm^{-3}

12.0 cm^3 is half the neutralisation volume, so pH = pK_a = 4.20 and $K_a = 10^{-4.20} = 6.3 \times 10^{-5}$ mol dm^{-3}.

QUICK CHECK

Which mixture is a buffer?

- hydrochloric acid and sodium chloride
- sodium hydroxide and sodium chloride
- ethanoic acid and sodium hydroxide in a 1 : 1 mole ratio
- ammonia and ammonium chloride

A buffer needs a weak acid or weak base together with its conjugate partner. Ethanoic acid and NaOH in a 1 : 1 ratio react completely, leaving only sodium ethanoate; with excess ethanoic acid the mixture would be a buffer.

QUICK CHECK

Why is the enthalpy change of neutralisation of ethanoic acid with NaOH slightly less exothermic than that of HCl with NaOH?

- Ethanoic acid forms less water per mole
- Energy is used to dissociate the weak acid during the reaction
- Ethanoate ions react with water exothermically
- Ethanoic acid has two acidic hydrogens

Both reactions form one mole of water per mole of acid, but ethanoic acid is mostly undissociated, and breaking the O–H bond to release its H^+ takes in some energy.